

A Learned Latent Algebra for Physical Systems Closed Under Spatial Composition and Time Evolution*

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Abstract

We ask whether the latent representation of a many-particle physical system can support a *compositional algebra* that is closed under both spatial composition and time evolution: encoding subsystems, combining them with a learned operator, and evolving the result should agree with first evolving the parts and then combining. Using a translation-invariant graph encoder, a symmetric learned combine operator $\oplus$, a learned one-step evolve operator $\mathcal{E}$, and density and velocity decoders trained against soft particle ground truth, we report three findings on two-dimensional rigid-body scenes. (i) A sliced-Wasserstein density loss is the lever that makes the rest possible, replacing a long-standing pixel-MSE blur ceiling at RMSE 0.190 with 0.114 at 64 particles and producing an embedding on which eight unrelated physical observables are linearly readable at $R^2 \geq 0.89$. (ii) Two paired-union losses on 128-particle scenes — a Wasserstein density term on $\text{enc}(A \cup B)$ and a combine-anchor term tying $\oplus(\text{enc}A, \text{enc}B)$ to $\text{enc}(A \cup B)$ — reduce 128-particle out-of-distribution density RMSE from 0.188 to 0.116 direct, and from 0.207 to 0.142 via combine, without degrading the in-distribution numbers. (iii) Adding a single commutativity loss $\|\mathcal{E}(\oplus(a, b)) - \oplus(\mathcal{E}(a), \mathcal{E}(b))\|^2$ drives test-time compose-evolve commutativity from cosine 0.86–0.99 down to ≥ 0.989 at horizons up to eight evolution steps, while density quality continues to improve (64p RMSE 0.093, 128p OOD 0.110). The algebra is approximately closed under both axes simultaneously. The same architecture transfers without modification across thermodynamic regimes: training on a Lennard-Jones fluid ($T^* = 1.0$) and gas ($T^* = 2.0$) yields density RMSE of 0.220 and 0.179 respectively, with velocity RMSE (0.314, 0.346) actually below the rigid-body baseline (0.400). Limitations remain: a 256-particle regime we never trained on, density reconstruction below the rigid-body baseline on the LJ fluid, and compositional and commutativity tests on LJ checkpoints not yet measured at test time.

1 Introduction

A practical lesson from the last decade of representation learning is that compositional structure in the embedding pays off downstream. Word embeddings carry analogies linearly [Mikolov et al.(2013)]; image features compose into scenes [Deng et al.(2009)]; language models exhibit predictable compositional scaling [Kaplan et al.(2020)]. In the physical sciences, an analogous question is whether a learned representation of a physical system can support algebraic operations that match the physics: combining two subsystems should produce a representation of the union, and evolving the combined representation in time should match what would have been obtained by evolving the parts separately and recombining.

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Geometric deep learning supplies one half of the answer in the form of equivariant networks [Cohen and Welling(2016), Thomas et al.(2018), Satorras et al.(2021)] and graph-based simulators [Battaglia et al.(2018), Sanchez-Gonzalez et al.(2020), Bronstein et al.(2021)]. Neural operators [Li et al.(2021), Kovachki et al.(2023)] learn maps between function spaces. None of these directly address whether the embedding itself — the vector that one would write to disk — supports a closed algebra over both *compose* (spatial union) and *evolve* (time advance). The closure question matters because closure is what allows reasoning at the level of the representation: a substrate for hierarchical simulation, multi-scale composition, and the kind of symbolic-algebraic operations on learned vectors that have proven productive in language and vision.

This paper is a progress report on that question for two-dimensional rigid-body scenes. The setup is deliberately simple: 64 particles drawn from 5 rigid objects (circles and rectangles) under gravity and elastic collisions. The architecture (Section 2) has a translation-invariant graph encoder, a symmetric learned combine operator $\oplus$, an evolve operator $\mathcal{E}$ with an analytical centre-of-mass update and a learned residual on the embedding, and decoders for density and velocity on a 32×32 grid. Training imposes path-independence between hierarchical decompositions of a particle set and the direct encoding of the union, plus a sliced-Wasserstein loss on density that we found to be central.

We organise our findings as a path from a strong density loss to algebra closure under both operators.

1. **Sliced Wasserstein replaces the MSE blur ceiling.** Pixel MSE on a density grid penalises mass at the wrong location at the same rate as no mass at all; the loss-optimal response is to smear. Replacing pixel MSE with a sliced-Wasserstein density loss [Rabin et al.(2011), Kolouri et al.(2019)] drops density RMSE from 0.190 to 0.114 at 64 particles, below the MSE oracle ceiling at ≈ 0.125 (Section 3).
2. **Multi-size paired-union supervision unblocks scene-size out-of-distribution generalisation.** Adding two paired-union losses on 128-particle unions of 64-particle batch samples — one Wasserstein density term and one combine-anchor term tying $\oplus(\text{enc}(A), \text{enc}(B))$ to $\text{enc}(A \cup B)$ — reduces 128-particle direct-encode density RMSE from 0.188 to 0.116 (−38%) and via-combine RMSE from 0.207 to 0.142 (−32%), with in-distribution 64p density slightly improving from 0.114 to 0.097 (Section 4).
3. **An explicit compose–evolve commutativity loss closes the algebra under time evolution.** Even before any explicit training, the path-independence anchor produces approximate commutativity: $\oplus \circ \mathcal{E} \approx \mathcal{E} \circ \oplus$ holds at cosine 0.99 for one step, decaying to 0.86 at eight steps. Adding a single MSE term between the two paths drives commutativity to cosine ≥ 0.989 out to $k = 8$, an order-of-magnitude reduction in MSE at every horizon. The algebra is closed under both axes within the trained regime (Section 5).

We are explicit about what is *not* established. The compose–evolve closure is demonstrated in the embedding, not in decoded observables: it says that paths through the algebra agree, not that the prediction at $k = 8$ steps is physically accurate. A 256-particle regime, unseen during training, remains hard (Section 8). Velocity reconstruction lags density on rigid bodies. The Lennard-Jones cross-regime results in Section 7 report density and velocity RMSE only; the compositional E1–E4 suite and commutativity tests on the LJ checkpoints are pending.

2 Method

A scene is a set of N particles $\{(x_i, v_i, m_i)\}_{i=1}^N \subset \mathbb{R}^2 \times \mathbb{R}^2 \times \mathbb{R}_{>0}$. We write the encoded state as a pair $(e, \phi) \in \mathbb{R}^d \times \mathbb{R}^5$, where e is a learned embedding and $\phi = (\bar{x}, \bar{v}, M)$ is an exact frame state with centre-of-mass position $\bar{x} \in \mathbb{R}^2$, centre-of-mass velocity $\bar{v} \in \mathbb{R}^2$, and total mass $M \in \mathbb{R}_{>0}$. The frame is computed analytically; the embedding is the output of a graph network on translation-invariant per-particle features.

2.1 Encoder, combine, evolve

Each particle is described by translation-invariant features $f_i = (x_i - \bar{x}, v_i - \bar{v}, m_i) \in \mathbb{R}^5$. A small MLP lifts f_i to a per-particle vector $h_i^{(0)} \in \mathbb{R}^d$, and we run two rounds of k -nearest-neighbour message passing in the CoM-relative geometry with $k = 8$:

$$h_i^{(\ell+1)} = h_i^{(\ell)} + \text{MLP}_u \left(h_i^{(\ell)}, \sum_{j \in \mathcal{N}_k(i)} \text{MLP}_e(h_i^{(\ell)}, h_j^{(\ell)}, \|x_i - x_j\|) \right), \quad (1)$$

followed by a LayerNorm. The embedding is a sum-pool plus a final MLP that takes $\log N$ as a side feature:

$$e = \text{LayerNorm} \left(\text{MLP}_p \left(\sum_i h_i^{(L)}, \log N \right) \right) \in \mathbb{R}^{128}. \quad (2)$$

The combine operator $\oplus : (\mathbb{R}^d \times \mathbb{R}^5)^2 \rightarrow \mathbb{R}^d \times \mathbb{R}^5$ is symmetric. The frame is closed-form: $\bar{x}_{a \oplus b} = (M_a \bar{x}_a + M_b \bar{x}_b) / (M_a + M_b)$ (analogously for $\bar{v}$ and M). The embedding is

$$e_{a \oplus b} = \text{LayerNorm}(\text{MLP}_c(e_a + e_b, e_a \odot e_b, |e_a - e_b|, \phi_{ab})), \quad (3)$$

where $\odot$ is elementwise product and $\phi_{ab} \in \mathbb{R}^4$ supplies inter-group context $(\|\bar{x}_a - \bar{x}_b\|, \|\bar{v}_a - \bar{v}_b\|, M_a + M_b, |M_a - M_b|)$. The triple $(e_a + e_b, e_a \odot e_b, |e_a - e_b|)$ is symmetric in a, b , so commutativity $a \oplus b = b \oplus a$ is exact by construction.

The evolve operator $\mathcal{E}$ updates the frame analytically by drift, $\bar{x} \mapsto \bar{x} + \Delta t \bar{v}$, with a learned residual MLP, and updates the embedding by a residual MLP followed by a learned projector $P : \mathbb{R}^d \rightarrow \mathbb{R}^d$ that pulls the result onto the valid manifold:

$$e_{t+\Delta t} = P(e_t + \text{MLP}_e(e_t)). \quad (4)$$

The same projector P is applied after every combine.

2.2 Decoders and ground truth

The density decoder projects (e, ϕ) to a 4×4 feature map and applies three transposed convolutions to produce a 32×32 probability map via sigmoid; the velocity decoder is identical with two output channels and no activation. The ground-truth density is a Gaussian kernel sum, $\rho_*(x) = \sum_i m_i \exp(-\|x - x_i\|^2 / 2\sigma^2)$ with $\sigma = 0.4$, evaluated on a 32×32 grid in a canonical frame defined by $\arg(M_2)/2$, where $M_2 = \sum_i m_i (x_i - \bar{x})^2 \in \mathbb{C}$ is the quadrupole moment.

2.3 Path-independence as a global anchor

Given a particle set S , a *decomposition tree* T has leaves that partition S . The tree-encoded embedding is obtained by encoding each leaf and combining bottom-up. Path-independence requires this to agree with the direct encoding:

$$\mathcal{L}_{\text{anchor}}(S) = \mathbb{E}_{T \sim \mathcal{T}(S)} \|e_T(S) - \text{enc}(S)\|_2^2. \quad (5)$$

A complementary dispersion term penalises tree-to-tree variance. Together they make the algebra approximately associative: if all trees of S embed to the same point, then any two trees do.

2.4 Phase A and Phase J training

Training proceeds in two phases. Phase A (warmup) trains the encoder, frame, structural readouts, density decoder, and velocity decoder on direct encodings only. Phase J (joint) re-enables every loss simultaneously: anchor, dispersion, density Wasserstein, velocity MSE, observable MSE, multi-step latent and observable evolution, conservation, micro-vs-macro consistency, energy consistency, and the additional terms introduced in Sections 4–5. A loss normaliser maintains a running max per term so that scales remain commensurable. The full configuration uses 1024 training trajectories of 64 particles each, 30 time steps per trajectory, $\Delta t = 0.005$, on a single RTX 3090 Ti for roughly twenty hours.

3 Sliced Wasserstein for density

Density reconstruction had been stuck at RMSE ≈ 0.19 for several iterations of the architecture (point-query MLP, spatial-broadcast with Fourier features, transposed convolution, doubled embedding dimension). Diagnostic baselines clarified the situation: a constant predictor (the dataset-mean density) achieves RMSE 0.134, and an oracle particle-to-MLP decoder seeing actual particle locations achieves 0.125. The model’s 0.146–0.190 band sat between these endpoints; capacity moved it within the band but never below the oracle ceiling.

The cause is structural. Pixel MSE between a predicted density $\hat{\rho}$ and target ρ_* is local: a unit of mass placed correctly at p_0 and a unit of mass placed at any other location $p \neq p_0$ contribute identically, regardless of $\|p - p_0\|$. Faced with uncertainty about object location, the loss-minimising response is to smear mass uniformly over the plausible support, since the resulting per-pixel error is smaller than concentrating in the wrong place.

The sliced Wasserstein-1 distance between two densities $\hat{\rho}, \rho_*$ on $\mathbb{R}^2$ is the expectation, over uniform unit directions $\theta \in S^1$, of the 1D Wasserstein-1 distance between their projections:

$$\text{SW}_1(\hat{\rho}, \rho_*) = \mathbb{E}_{\theta \sim \text{Unif}(S^1)} \int_{\mathbb{R}} |F_{\theta\# \hat{\rho}}(t) - F_{\theta\# \rho_*}(t)| dt. \quad (6)$$

We approximate the expectation with $K = 64$ random directions per minibatch and discretise the 1D integral by sorting projected grid coordinates and integrating the absolute CDF gap; gradients flow through the sort. Mass at the wrong location now costs $O(\|p - p_0\|)$, not $O(1)$, so smearing is no longer an attractor [Rabin et al.(2011), Deshpande et al.(2018), Kolouri et al.(2019), Villani(2008)]. With architecture, capacity, and dataset held fixed, replacing pixel MSE with SW_1 on the density grid yields the change in Table 1.

The 0.114 figure is below the MSE oracle because SW_1 accepts a sharper prediction at the cost of a small mis-centring that pixel MSE would punish heavily. Linear probes on eight held-out observables — radius of gyration, mean pairwise distance, n -cluster count, total angular speed, ψ_6 ,

Table 1: Density reconstruction at 64 particles, 5 rigid objects, 128-dim embedding, 1024 training trajectories. SW is the only change between rows; all other losses are identical.

Loss on density grid	Density RMSE	Notes
Constant predictor (dataset mean)	0.134	target-side floor
Oracle (particles $\rightarrow$ MLP)	0.125	MSE-attainable ceiling
Pixel MSE	0.190	long-standing blur ceiling
Sliced Wasserstein-1 (this work)	0.114	below the MSE oracle

internal kinetic energy, bounding radius, spatial standard deviation — all attain $R^2 \geq 0.89$. The SW₁ loss does not merely improve visual reconstruction; it improves the embedding’s information content as measured by independent linear probes.

This is the foundation we build on. Every subsequent result uses pure-SW density as the in-distribution baseline.

4 Multi-size training closes the OOD scene-size gap

The clearest limitation of the SW-only checkpoint was scene-size out-of-distribution generalisation. A 128-particle scene encoded directly by the 64p-trained encoder reached density RMSE 0.188; composing two 64-particle scenes via $\oplus$ landed at 0.207. Both were $\sim 65\%$ worse than the in-distribution baseline. The encoder, not the algebra, was the floor.

We add two paired-union losses to Phase J. For each minibatch we randomly partition the B samples of 64-particle scenes into pairs (A_i, B_i) , place B_i six units offset from the origin, and form the union $A_i \cup B_i$ with 128 particles.

Density on the union. A sliced-Wasserstein density loss between $\text{enc}(A \cup B)$ ’s decoded density and the Gaussian-kernel ground truth of the 128-particle union (weight 2.0):

$$\mathcal{L}_{\text{dens-2x}} = \text{SW}_1(D(\text{enc}(A \cup B)), \rho_*(A \cup B)). \quad (7)$$

This exposes the encoder to 128-particle inputs.

Combine-anchor. An MSE between $\oplus(\text{enc}(A), \text{enc}(B))$ and $\text{enc}(A \cup B)$ (weight 1.0):

$$\mathcal{L}_{\text{combine-anchor}} = \|P(\oplus(\text{enc}(A), \text{enc}(B))) - \text{sg}[\text{enc}(A \cup B)]\|_2^2, \quad (8)$$

with stop-gradient on the target. This supplies $\oplus$ with direct compositional supervision, anchoring the combined representation to the directly-encoded union rather than only to the indirect path-independence target.

We retrain from scratch (`--no-resume`) with all original Phase J losses plus these two terms and evaluate on the same eval suite as before. Table 2 compares pure-SW (the original headline checkpoint of Section 3) and multi-size (pure-SW plus the two paired-union terms).

The OOD scene-size gap that was the principal open issue of the pure-SW checkpoint is largely closed. With 128-particle paired-union supervision, both direct and via-combine density at 128 particles drop to within striking distance of the in-distribution baseline. Encoding had been the bottleneck: once the encoder sees 128-particle inputs, it generalises cleanly. The 256-particle regime remains hard because we still never train on it.

Table 2: Effect of paired-union supervision on in- and out-of-distribution metrics. “Direct” = encoder applied to the full particle set; “via combine” = halves encoded separately and joined via $\oplus$. Lower is better.

Metric	Pure-SW	Multi-size	Δ
64p density RMSE	0.114	0.097	−15%
64p velocity RMSE	0.372	0.400	+8%
Particle-NN velocity RMSE	0.703	0.736	+5%
Phase-space W_1	0.146	0.135	−8%
E2 32+32 $\rightarrow$ 64 latent MSE	0.90	1.10	+22%
E2 density via combine	0.130	0.131	flat
E2 density via direct	0.111	0.094	−15%
E3 128p direct OOD density	0.188	0.116	−38%
E3 128p via combine density	0.207	0.142	−32%
E4 256p balanced density	0.253	0.254	flat
E4 256p chain density	0.254	0.254	flat

The price is a slight regression in velocity, both pixelwise ($0.372 \rightarrow 0.400$) and at particle locations ($0.703 \rightarrow 0.736$). The two new losses concentrate on density only; we did not add a velocity-side analogue. The latent MSE in E2 grows somewhat ($0.90 \rightarrow 1.10$) because the geometry of the embedding has shifted to accommodate the 128p regime; decoded densities are unchanged or improved.

5 Compose–evolve commutativity

The algebra has so far been constrained on the spatial axis: combining the encodings of subsets and demanding agreement with the union. The natural next constraint is closure on the time axis. The encoded state (e, ϕ) supports a one-step evolution operator $\mathcal{E}$. Closure under both compose and evolve says that

$$\mathcal{E}(\oplus(a, b)) \stackrel{?}{=} \oplus(\mathcal{E}(a), \mathcal{E}(b)), \quad (9)$$

which we will call *compose–evolve commutativity*. This is the algebraic property of a tensor product on representations of a Lie semigroup: combining and evolving commute [Villani(2008)].

5.1 Loss

We add a third paired-union loss to Phase J (weight 1.0):

$$\mathcal{L}_{\text{compose–evolve}} = \|P(\mathcal{E}(\oplus(\text{enc}A, \text{enc}B))) - P(\oplus(\mathcal{E}(\text{enc}A), \mathcal{E}(\text{enc}B)))\|_2^2. \quad (10)$$

Empirically, this loss starts at $\approx 7 \times 10^{-4}$ at the beginning of Phase J — already small, suggesting commutativity is partially implicit in the path-independence anchor — and explicit training drives it lower.

Table 3: Compose–evolve commutativity at test time. MSE is between $\mathcal{E}^k(\oplus(a, b))$ and $\oplus(\mathcal{E}^k a, \mathcal{E}^k b)$ on validation pairs; cosine similarity is between the same vectors. Pure-SW had no commutativity training; compose-evolve adds the explicit MSE term.

Horizon k	Pure-SW MSE	Compose-evolve MSE	Pure-SW cos	Compose-evolve cos
1	0.0110	0.0008	0.9882	0.9998
2	0.0292	0.0025	0.9678	0.9994
4	0.0675	0.0087	0.9278	0.9979
8	0.1424	0.0474	0.8593	0.9890

Table 4: Density RMSE across the three checkpoints. Each adds losses to the previous; nothing is removed. Compose-evolve continues to improve every density metric tracked while driving commutativity to ≥ 0.989 at $k \leq 8$.

Metric	Pure-SW	Multi-size	Compose-evolve
64p density RMSE	0.114	0.097	0.093
E3 128p direct OOD density	0.188	0.116	0.110
E3 128p via combine density	0.207	0.142	0.135

5.2 Test-time commutativity

We measure compose–evolve commutativity on 200 independent pairs of validation scenes (A, B) , each pair offset by six spatial units. For horizon $k \in \{1, 2, 4, 8\}$ we compute

$$e_k^{(1)} = \mathcal{E}^k(\oplus(\text{enc}A, \text{enc}B)), \quad (11)$$

$$e_k^{(2)} = \oplus(\mathcal{E}^k \text{enc}A, \mathcal{E}^k \text{enc}B), \quad (12)$$

and report $\|e_k^{(1)} - e_k^{(2)}\|^2$ and the cosine similarity between them. Table 3 compares the pure-SW checkpoint (no compose–evolve loss) against the compose–evolve checkpoint, which adds all three paired-union terms.

Two observations stand out. First, even without explicit training, the pure-SW checkpoint shows non-trivial commutativity: cosine 0.99 at $k = 1$, decaying to 0.86 at $k = 8$. The path-independence anchor does most of the work; closure under time evolution is partially implicit in the algebra constraint imposed on the spatial axis. Second, an explicit loss on the commutativity gap drives test-time MSE down by an order of magnitude at every horizon and cosine to ≥ 0.989 out to eight evolution steps. Within the trained regime the algebra is closed under both operators to within $\sim 1^\circ$.

What this does *not* say: it says nothing about whether $\mathcal{E}^k$ produces a physically accurate evolution at $k = 8$. We are measuring agreement *between two paths through the algebra*, not agreement of either path with a ground-truth simulator. Compose–evolve closure is an algebraic consistency property; it is necessary but not sufficient for accurate long-horizon prediction.

5.3 Effect on density and on the rest of the suite

Adding compose–evolve to multi-size does not regress the other metrics; it slightly improves them. Table 4 tracks 64p and 128p density RMSE across the three checkpoints.

The compose-evolve term acts as a mild regulariser: it constrains $\oplus$ and $\mathcal{E}$ to be jointly consistent, which appears to slightly shrink the configuration space of valid embeddings without forcing any particular density trade-off.

6 Compositional generalisation, end to end

We summarise here the full battery of compositional tests on the final compose-evolve checkpoint, with the intermediate and pure-SW numbers carried over for context. Definitions of the four test types follow the previous draft [Rabin et al.(2011)] and are reproduced in shortened form for completeness.

E1 — Out-of-distribution tree shapes. Build a decomposition tree of a 64-particle scene with group sizes the model never saw during training. Tree shapes with $\{3, 5, 7, 12, 24\}$ groups are out-of-distribution; $\{4\}$ is in-distribution. Compare the tree-encoded embedding to the direct encoding via latent MSE and cosine similarity. Pure-SW achieves cosine 0.50 at 4 groups (the in-distribution case), 0.62 at 3 groups (smaller and bigger groups need fewer combine steps and accumulate less drift), and 0.29 at 24 groups. Degradation is graceful, not catastrophic.

E2 — Subset composition. Split a 64-particle scene into two halves of 32, encode each, combine, compare. One combine step adds $\sim 17\%$ to density RMSE in pure-SW (0.130 via combine vs 0.111 direct). Multi-size and compose-evolve do not improve this gap: the pure-SW value 0.130 is roughly preserved at 0.131, while the direct number drops because the encoder is stronger. The single-step combine cost is real and quantifiable.

E3 — Cross-scene union. The headline OOD test, reported in full above. Pure-SW 0.207 (combine) vs 0.188 (direct); compose-evolve 0.135 (combine) vs 0.110 (direct).

E4 — Associativity. Take four 64-particle scenes and form their union via balanced $((AB)(CD))$ and chain $((((AB)C)D))$ trees. Pure-SW: latent MSE 0.32, cosine 0.71 between the two trees, decoded densities differing by 0.001 (RMSE 0.253 vs 0.254). Multi-size and compose-evolve are flat at ≈ 0.254 . Two genuinely different tree shapes over 256 particles produce nearly identical embeddings and identical decoded densities. Associativity is the cleanest positive result on the spatial axis: the combine operator behaves like an associative monoid in distribution, despite no architectural constraint enforcing it. The absolute number ≈ 0.254 remains poor because 256 particles is OOD.

7 Cross-regime transfer: rigid bodies, Lennard-Jones fluid, and Lennard-Jones gas

The closure results above are on a single physical domain (rigid bodies under gravity and elastic collisions). A natural test of the framework is whether the same architecture and the same loss inventory transfer to a qualitatively different physics — continuous pair interactions, no rigid structure, temperature-dependent phase behaviour — without any architectural change.

We train two additional checkpoints on the Lennard-Jones (LJ) system. The simulator integrates $N = 64$ particles in a 2D box at reduced density $\rho^* = 0.7$ under the standard 12-6 pair potential, thermalised by a Langevin thermostat at reduced temperature T^* . Two regimes are reported: $T^* = 1.0$ (a fluid with significant clustering and voids) and $T^* = 2.0$ (a gas with near-uniform

Table 5: Density and velocity reconstruction across thermodynamic regimes. All numbers are at 64 particles, in-distribution evaluation. Same V4 architecture, same loss inventory, same 1024 training trajectories. Only the simulator and (for LJ) the temperature change.

Regime	Density RMSE	Velocity RMSE
Rigid bodies (in-dist 64p)	0.093	0.400
LJ fluid ($T^* = 1.0$)	0.220	0.314
LJ gas ($T^* = 2.0$)	0.179	0.346

density). Training uses the same 1024 trajectories, the same V4 architecture (translation-invariant graph encoder, learned $\oplus$, learned $\mathcal{E}$, density and velocity decoders), and the same Phase A / Phase J loss schedule, including the multi-size paired-union and compose–evolve commutativity terms. The only differences from the rigid-body configuration are `sim_type=lj` and the temperature. Wall-clock times were 27.7 hours for $T^* = 1.0$ and approximately 14.4 hours for $T^* = 2.0$ on a single RTX 3090 Ti.

Table 5 compares the headline reconstruction metrics across the three regimes.

Two interpretive points stand out.

Density follows the structure. Density reconstruction is easiest on rigid bodies, where mass is concentrated on a small number of compact objects that the encoder can localise. It is hardest on the LJ fluid, where the particle distribution is intermediate — diffuse but with emergent clusters and voids that the decoder must reproduce. The LJ gas is in between: more diffuse than rigid but more spatially uniform than the fluid, and the more uniform target is easier to reconstruct than the structured fluid. The ordering rigid < gas < fluid for density RMSE is what one would predict from the spatial complexity of the underlying mass distribution. Absolute LJ numbers are roughly twice the rigid-body baseline, which is consistent with continuous LJ density being intrinsically harder to capture on a 32×32 grid than five rigid blobs.

Velocity is better on LJ than rigid. Velocity RMSE on rigid bodies is 0.400; on both LJ regimes it is lower (0.314, 0.346). On rigid bodies, particles within a single object share a velocity, so velocity content is dominated by a small number of object-level rigid motions; the encoder need not preserve fine-grained per-particle velocity to do well on object-level dynamics. On LJ, every particle has its own thermal velocity, and the velocity field carries genuine per-particle information that the encoder is forced to represent if the velocity decoder is to fit the ground truth. The encoder learns more useful velocity structure when the physics demands it. The rigid-body velocity gap noted in Section 8 thus has a domain-specific component: rigid-body velocity is partly a representation choice rather than a fundamental modelling limit.

What this is and is not. The framework transfers across thermodynamic regimes without architectural changes. This is the cross-domain claim that the original draft of this paper carried as future work. What we have not measured on the LJ checkpoints is the compositional E1–E4 suite of Section 6 or the test-time compose–evolve commutativity of Section 5. The training-time commutativity loss $\mathcal{L}_{\text{compose–evolve}}$ ended at $\approx 2 \times 10^{-4}$ for $T^* = 1.0$ and was at $\approx 4 \times 10^{-4}$ mid-training for $T^* = 2.0$, both within the same range as the converged rigid-body checkpoint, which is consistent with the property holding during training; but the test-time numbers are pending and we make no claim of LJ closure on the time axis.

8 Limitations and negative results

We catalogue the failure modes honestly because they sharpen what the positive results actually show.

256-particle scenes are still hard. Multi-size training on paired 128-particle unions does not extrapolate to 256 particles: E4 density stays flat at ≈ 0.254 across all three checkpoints. The natural fix — training on triples or quadruples of 64-particle samples — has not yet been run. The pattern is consistent with our earlier reading of the OOD scene-size effect: the encoder can interpolate within the trained range but does not extrapolate outside it.

Velocity reconstruction lags density. The fair particle-NN velocity RMSE rises from 0.703 (pure-SW) to 0.736 (multi-size) and stays in the 0.73 band with compose-evolve. The new losses target density and embedding, not velocity. We have no velocity-side analogue of the SW improvement: an earlier attempt at a 4D phase-space sliced Wasserstein loss regressed both density (to 0.168) and velocity (particle-NN 0.787), because spatial supervision was diluted across 4 projection dimensions. Phase-space SW is retained only as a low-weight auxiliary, not a primary loss.

Algebra closure is not physical accuracy. Compose-evolve commutativity at horizon $k = 8$ says that two paths through the algebra agree, not that either path’s prediction is physically accurate at $k = 8$. The decoded density at $k = 8$ has error from the $\mathcal{E}$ operator’s residual, which is not measured by our commutativity test.

LJ compositional and commutativity tests are pending. Section 7 reports LJ cross-regime density and velocity RMSE only. The compositional E1–E4 evaluation (out-of-distribution tree shapes, subset composition, cross-scene union, associativity) and the test-time compose-evolve commutativity sweep have not been run on the LJ checkpoints. The training-time commutativity loss was at the same floor as the converged rigid checkpoint, suggesting the property held during training, but we do not claim test-time LJ closure on either axis.

Several over-engineered embeddings did worse, recorded for posterity. Earlier iterations used 34-dimensional analytical multipole moments with a binomial-shift combine, 9-dimensional PCA-canonical features, and split position/velocity channels. Each was theoretically attractive and each was beaten by the simple translation-invariant graph encoder reported here. PCA orientations were unstable near isotropic configurations; mixing analytical moments with learned scalars destroyed compositionality; channel splits over-constrained the encoder. Prescribing the meaning of embedding dimensions traded composability for representational capacity, and the trade was not in the prescriber’s favour.

9 Related work

The architectural backbone is a graph network [Battaglia et al.(2018), Sanchez-Gonzalez et al.(2020)] with translation-invariant inputs, in the lineage of equivariant networks [Cohen and Welling(2016), Thomas et al.(2018), Satorras et al.(2021)] and the broader programme of geometric deep learning [Bronstein et al.(2021)]. The decoder is a deconvolution from a small spatial seed, standard in object-centric and slot-based models [Watters et al.(2019), Locatello et al.(2020), Greff et al.(2020)].

Time evolution in embedding space connects to neural ODEs [Chen et al.(2018)] and, at population level, neural operators [Li et al.(2021), Kovachki et al.(2023)].

The density loss is sliced Wasserstein [Rabin et al.(2011), Kolouri et al.(2019), Deshpande et al.(2018)] with optimal-transport theory in [Villani(2008)]. The closest conceptual neighbours on the algebra side are works that learn coarse-graining or renormalisation for physical systems [Wilson and Kogut(1974)] and graph-network simulators [Sanchez-Gonzalez et al.(2020)]. Compositionality of learned representations under group operations has been studied for image-domain transformations [Cohen and Welling(2016)] and for object-centric scenes [Locatello et al.(2020), Greff et al.(2020)]; we are not aware of prior work that directly tests closure under both spatial composition and time evolution in an embedding learned from a particle simulator. Concurrent work on coarse-grained molecular simulation with learned message-passing [Sanchez-Gonzalez et al.(2020)] shares the simulator-side motivation; our concern is the algebra of the embedding, not the simulator’s rollout fidelity.

10 Discussion and future work

A working hypothesis at the start of this project was that a learned latent algebra could describe a physical system, with the embedding closed under both compose and evolve. The progress reported here is consistent with that hypothesis at the trained regime: with a sliced-Wasserstein density loss to give the encoder a non-degenerate target, paired-union supervision to extend the encoder’s range across scene sizes, and a single MSE term tying $\mathcal{E} \circ \oplus$ to $\oplus \circ \mathcal{E}$, the algebra becomes approximately closed on both axes within the regime it was trained on. The 128-particle out-of-distribution gap that was the main open issue of the original draft is largely closed; compose–evolve commutativity, which was emergent at cosine ≈ 0.86 – 0.99 from the path-independence anchor alone, is now ≥ 0.989 out to eight evolution steps.

The picture has natural next steps.

1. **Extend the trained regime.** Train on triples and quadruples of 64-particle samples (i.e. 192, 256, 320 particles) to push the multi-size frontier. The pattern in Section 4 suggests 256p will follow 128p down once it is in distribution.
2. **Run the compositional and commutativity suites on LJ.** Section 7 establishes density and velocity transfer across rigid, LJ fluid, and LJ gas, but does not measure E1–E4 or the test-time compose–evolve cosine sweep on the LJ checkpoints. These are the natural next evaluations and require no further training.
3. **Complete the LJ phase diagram.** A $T^* = 0.3$ solid LJ run would join the fluid and gas checkpoints to span the three canonical thermodynamic regimes. Adding a third domain — the 2D Ising model — would test transfer to a discrete-spin system rather than another continuous-particle one, and would let the same closure tests be run on a renormalisation-group-style coarse graining.
4. **Velocity-side analogue of the SW lever.** Density quality is now well above velocity quality. The straightforward thing to try is a SW loss on the joint position-momentum distribution restricted to the spatial slice (the 2D marginal), or a transport-based loss on the velocity-weighted density. The phase-space attempt failed because of dimension dilution; a hybrid that keeps spatial supervision sharp may not.
5. **Decoded-space commutativity at long horizon.** Compose–evolve currently measures agreement in the embedding. The natural follow-on is to measure agreement in decoded density

and velocity at $k = 8, 16, 32$ steps, against a ground-truth simulator. This separates algebra closure from prediction accuracy.

6. **Downstream prediction tasks.** The eight linear-probe observables of Section 3 were chosen to be unsupervised diagnostics of embedding content, not endpoints. A more demanding test of the algebra is whether the embedding supports nontrivial downstream prediction — phase-transition detection on LJ, energy and pressure regression, lifetime prediction of metastable configurations — using only linear or shallow heads on the frozen embedding. Larger-scale runs (the local single-3090 box has been the bandwidth ceiling on dataset size; cloud-GPU migration is in progress) will make these tractable.

The longer-term motivation is the substrate question: if a representation is reliably closed under compose and evolve, it can be used directly for hierarchical reasoning across scales, explicit composition of independent simulations, and the kind of latent algebra that has been so productive in language models [Mikolov et al.(2013), Kaplan et al.(2020)]. The present paper’s contribution is narrower: a recipe under which the algebra demonstrably exists on a single physical domain, an inventory of which losses produce which closure properties, and a careful catalogue of where it currently breaks.

Reproducibility. Numbers in Tables 1–4 are taken directly from the evaluation outputs of the `rigid.wasserstein_pure`, `rigid.multisize`, and `rigid.compose_evolve` checkpoints; numbers in Table 5 are from the `lj_T1.0` and `lj_T2.0` checkpoints, evaluated by `eval_lj.py` on 128 held-out trajectories. We report single-run point estimates and have not yet aggregated over seeds.

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